SNF-Bench++: Evaluating Fixed-Camera Video Generation Across Sustained Motion and Long Horizons

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***Abstract*—Long-running fixed-camera video generation requires stationary scene support and sustained, plausible motion in the intended region. Existing video quality and consistency measurements provide useful signals but do not by themselves establish whether generated motion has the right material, direction, or temporal evolution. This working manuscript specifies a measurement framework for text- and image-conditioned evaluation that separates support fidelity, material behavior, temporal continuity and human judgments. Its development draws on earlier SNF-Bench evaluations and a source-fixed image-to-video protocol. The independent journal confirmation set, experimental panel and human-alignment results remain to be collected; this draft makes no claim that the candidate measures have been validated.**

***Index Terms*—video generation, fixed camera, long horizon, benchmark, perceptual evaluation**

I. Introduction

FIXED-CAMERA animation appears in nature observation and sustained visual content. A plausible output must preserve stable support without freezing the intended moving material, while avoiding replay, drift and scene corruption. General-purpose video benchmarks already measure important aspects of quality and consistency [3]–[7]; a task-specific benchmark must demonstrate its additional value on shared inputs rather than assume those measures are blind to fixed-camera failure.

Long-horizon generators are now evaluated at minute scale [8]–[12], yet two failure modes of this setting are rewarded by measures that read a statistic in one direction. Activity measures reward any motion, including incoherent or boiling motion. Source-referenced fidelity measures reward any support that stays close to the source, including a pasted still image. A fixed-camera benchmark therefore needs measures that are two-sided, region-aware and validated against real footage.

We study a versioned evaluation protocol for text-to-video and image-to-video outputs. The contribution is an independently validated measurement and evaluation resource, not a new generation method. The original SNF-Bench preprint [13] and the source-fixed method-paper protocol are prior evidence; Appendix A states what is new here. Human and animal animation are possible separately validated extension tracks, not evidence for the present nature-flow core.

The contributions are:

- **A fixed-camera benchmark suite** for text- and image-conditioned generation at 60, 120 and 240 seconds, with a regime taxonomy, source-derived role maps, contamination-audited splits and a datasheet (Section III).
- **A measurement suite** that separates support fidelity, material behavior, continuity and two-sided realism, with explicit eligibility and abstention (Section IV).
- **A validation protocol** in which every measure must pass constructed adversaries, real-footage bands and blinded human judgments before admission (Section V).
- **An evaluation of published long-horizon generators** under one protocol, with per-regime results and findings (Sections VI and VII). [Pending: panel size and headline findings]

II. Related Work

Video-generation benchmarks. Multi-dimensional suites decompose quality into named dimensions and validate each against human preference [3]–[6], [14], [15]. WorldScore [7] adds source-anchored regional motion, and VMBench [16] and Video-Bench [17] target motion perception and judge models. VBench already reports a trade-off between temporal consistency and dynamic degree and keeps the two as separate dimensions [3], and WorldScore reports static and dynamic aggregates separately, measuring geometric drift by SLAM reprojection error [7]. The premise that consistency and motion must be reported separately is therefore established; what remains open in the fixed-camera, minute-scale setting is whether region-specific measures add information over these incumbents on the same videos, which Section V tests directly, including where incumbents already agree. These suites mostly evaluate clips of a few seconds and read each dimension in one direction.

Dynamics, time-lapse and physics. DEVIL [18] grades dynamics against the prompt and DIVE [19] rates image-to-video dynamics, both penalizing static outputs that would otherwise score well on consistency; ChronoMagic-Bench [20] evaluates metamorphic time-lapse content, and VideoPhy [21] and Physics-IQ [2] test physical plausibility, the latter against real continuations. None of these isolates stationary support from intended motion over minutes.

Motion and quality metrics. Distribution metrics compare generated and real videos [22], [23]. TRAJAN [24] scores single videos from point-track reconstruction. Quality-assessment datasets collect mean opinion scores for generated video [25], [26]. We evaluate such measures as comparators and candidate statistics inside role masks; their suitability for water, smoke and fire over minutes is measured, not assumed.

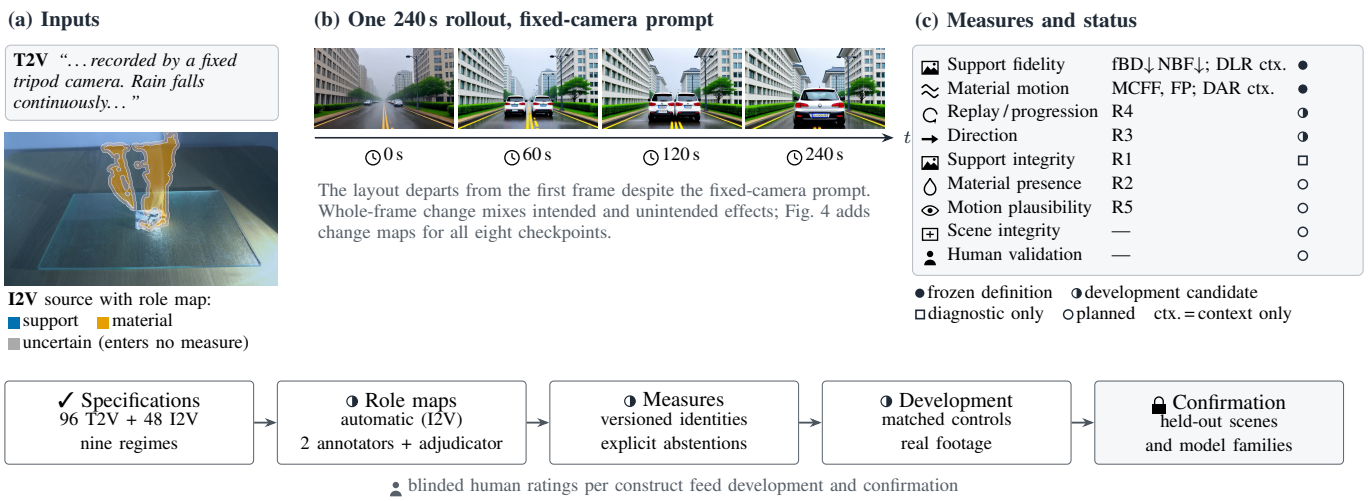

Fig. 1. **SNF-Bench++ at a glance.** (a) Each scene specifies a fixed viewpoint and names a natural process; I2V sources carry role maps; illustrated on a public fixed-camera scene, smoke rising from burning paper, with roles derived from that benchmark's motion masks (overlay pixels such as falling snow, also excluded, are absent here). (b) Frames of a historical output of a published checkpoint [1] under the common inference configuration for a fixed-camera prompt (first system alphabetically, first 240 s prompt; Fig. 4). (c) Measures by construct and validation status; no summed score. Bottom: the protocol, from specification to held-out confirmation. Image in (a): Physics-IQ benchmark [2], scene 0140, first frame, © 2024 DeepMind Technologies Limited, CC BY 4.0; resized, with roles overlaid. An interactive version accompanies the supplementary material (`interactive/measurement_map.html`).

Long-horizon and fixed-camera generation. Autoregressive and diffusion-forcing generators extend rollouts to minutes [1], [8]–[12]. Single-image cinemagraph methods [27], [28] animate a region while holding the rest still, which makes them a natural composite-type entrant.

Entrant policy. Every entrant is a published method evaluated as released. Steady-Forcing [29] is entered as its public preprint, and SceneTether [30] will be entered only once its preprint is public. Both are judged under the same rules as external methods, each as its single released configuration; no ablation or internal variant of either appears in any table. The final comparison and bibliography require verification at the journal freeze.

III. Benchmark Construction

Figure 1 summarizes the proposed workflow; its annotations, real-footage validation and held-out confirmation are still pending.

A. Task

The core task conditions on a fixed viewpoint and a specified natural process. The image-conditioned track uses a source image and prompt; the text-conditioned track has no common source image and requires its own partition and annotation policy.

B. Scene Suites and Taxonomy

Scenes are stratified by transport regime, the physical way material moves (for example channel flow, falling water, combustion and precipitation), and results are macro-averaged over regimes so that no regime dominates. Figure 2 shows the specified scene counts, not completed generations. The four water regimes together account for 42/96 text-conditioned and 21/48 image-conditioned specifications. The text-conditioned

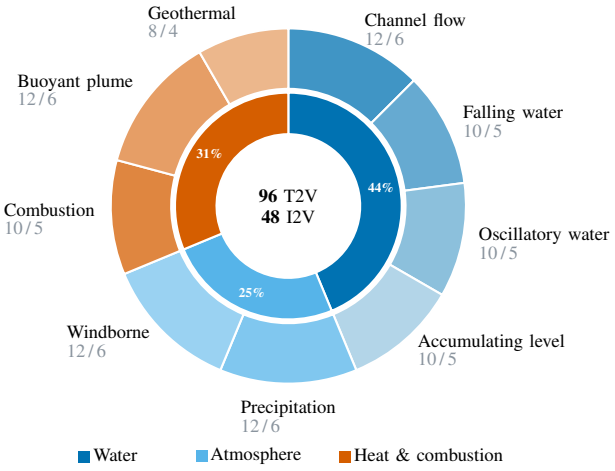

Fig. 2. **Specified scene composition.** Inner ring: medium family (display grouping); outer ring: transport regime, labelled with T2V / I2V scene counts. Each regime holds 8.3–12.5% of a track, but the four water regimes together hold 44% (42/96 and 21/48), so water is also reported as one group. Directions are declared for 69/96 T2V and 43/48 I2V scenes; they are text-derived candidates pending independent annotation. These are specifications, not completed renders, validated masks or rights clearance. Filterable list: `interactive/scene_explorer.html`.

suite specifies the same scenes at every horizon; every image-conditioned source is cut to 832×480, the native resolution of the evaluated base family; any entrant whose released pipeline resizes or re-encodes is recorded as such. [Pending: caption lengths, source resolution and provenance]

C. Splits and Contamination Audit

Existing source sets have informed method development and therefore are not an untouched journal confirmation cohort. Each confirmation source is checked against every development, training and earlier evaluation image by exact hash,

perceptual hash and geometric verification, and sources reused from earlier evaluations are reported as a separate seen-overlap stratum. [Pending: fresh rights-cleared confirmation set and its audit counts]

D. Role Maps

The planned role maps distinguish stable support, target material, moving subjects, overlays and uncertain regions, so that, for example, a moving boat and the water it floats on can be measured separately. Observable material directions are to be annotated as image-plane sectors, with uncertain and multi-direction labels. The planned protocol uses two independent annotators, reports agreement and adjudicates or marks disagreements uncertain. In the image-conditioned track, source-derived maps supply common regions across systems; the existing maps are not yet independently human-verified. The text-conditioned track has no common layout, so its planned regions require blinded per-output annotation and uncertainty reporting. [Pending: annotation procedure, reviewer independence, agreement statistics]

E. Horizons and Generation Protocol

The planned primary horizons are 60, 120 and 240 seconds, with a short control. Same-rollout prefixes and separately requested durations are distinct experimental units. Each entrant runs with its released code, weights and default sampler. Native long-video inference, documented extensions and wrapped short-clip models are separate row types and are never pooled. Seeds are fixed per track. Each output's resolution, frame rate and encoder are recorded, and scoring resamples to a common analysis rate. Differences that remain are reported as part of the row type rather than removed. Final sampling, source rights, source-family splits, annotation procedures and model configurations must be frozen before the confirmation analysis. [Pending: entrant table with release revision, row type, seeds and compute]

IV. Measurement Suite

Support fidelity includes source-relative appearance and geometric displacement, with registration failures reported. **Material behavior** separates activity, active area and direction where observable. **Continuity** targets replay and resets while retaining legitimate periodic flow. **Scene integrity** addresses missing or invented content beyond the measured support. Human ratings assess the corresponding observable attributes separately. Figure 3 shows which role each measure reads.

These are candidate axes, not an admitted headline score. A sharp static frame can preserve detail without motion; noise can increase activity; natural surf can exhibit periodicity. Frame-based displacement (fBD) remains a diagnostic, not a sufficient surrogate for source fidelity. A historical mean-colour field has a documented float32 accumulation defect and retains its original measurement identity.

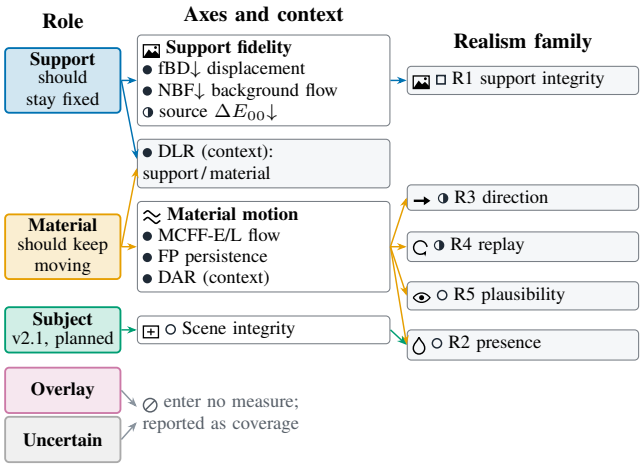

Fig. 3. **Illustrative role routing.** Lines show intended inputs, not complete computational dependencies: DLR compares roles and motion compensation can use support information. Scene integrity and subject handling are planned; R2 is material presence. Overlay and uncertain pixels are excluded from the current partitioned measures and reported as coverage. Realism measures are reported beside the axes and never summed with them. A downward arrow marks a directional support statistic, not an overall quality score; material activity and the contextual ratios require joint interpretation. ● frozen definition, construct validation pending; ◑ development candidate; □ diagnostic only; ○ planned. Interactive version: `interactive/measurement_map.html`.

A. Realism Family

Realism is a separate family of measures, reported beside support fidelity, material behavior and continuity, and never summed with them. Each measure names the single construct it targets. A measure that orders a statistic in one direction can reward an exploit: activity rewards incoherent motion, and closeness to the source rewards a pasted still image. Each realism statistic is therefore read relative to an explicit reference: real fixed-camera footage of the same medium where the construct allows, otherwise encoded controls.

- *Support integrity.* Successive-frame change in the support after drift and global-illumination compensation, read against real footage and against pass-through, pasted, grain-matched and feathered composites. It is a diagnostic: a valid fixed background can be still and a careful composite can look natural, so it does not by itself establish that a support was pasted or redrawn.
- *Material presence and location.* Whether the requested material, rather than another region or a subject, carries the motion. It is checked against source-conditioned material and subject annotations; for example, a moving boat over frozen water is not persistent water flow.
- *Direction.* The flow component along an image-plane direction that is annotated independently of the scene text, with unknown, ambiguous and multi-direction options, defined only where the direction is visible, and tested against time-reversed and ping-pong controls of matched magnitude.
- *Progression versus replay.* Multi-frame recurrence with seam and reset evidence, gated by active area. Detection and false-positive rates are reported separately, with natural periodic surf, fire and foliage as negatives.

- *Motion plausibility.* Flow, point-track and dynamic-texture statistics inside the dynamic role, compared with medium-level distributions of real footage with source-family uncertainty, and with established comparators on the same eligible inputs: TRAJAN per video [24], FVMD at set level only [23], and VBench-2.0 physics [5] with any adaptation recorded.

A real continuation of a source frame is one plausible realization, not the correct trajectory, so no measure compares pixels with it. No two-dimensional statistic establishes physical correctness, and none of these measures is claimed validated before Section V. A scientifically plausible cinemagraph can be a valid output; a composite counts as a defect only where the task or human raters say so.

Eligibility and abstention. A measure abstains when its region is empty or registration to the source fails, and the number of eligible videos is reported beside every value. Abstentions are never imputed, and registration failures are counted rather than dropped.

V. Metric Validation

The planned study tests paired interventions, natural periodic and occlusion controls, and real generated failures. Development and confirmation data are split by scene family and model family, keeping all crops, horizons, controls and seeds of a scene together. Thresholds and any metric-selection decisions are established on development data, with final performance assessed on held-out data under a preregistered analysis. Each metric must report eligibility, abstentions, false-positive behavior and uncertainty; response direction is separate from task desirability. No controlled intervention alone establishes perceived naturalness.

A. Targeted Controls

Each measure is tested only on the failures it targets, and for invariance to declared nuisances. It is never required to detect every corruption. For each measure, the targeted control families, the nuisances, the minimum eligible coverage, the comparison with a simple established measure and the primary effect-size threshold are fixed on development data before any confirmation score is inspected. Controls include:

- pass-through re-encodes;
- pasted, grain-matched, feathered and illumination-matched composites;
- time-reversed, ping-pong, exact and cross-faded loops;
- freezes;
- noise warps and boiling texture;
- real footage resampled to other playback speeds;
- unrelated subject motion.

Each control is built from the same decoded parent as its pass-through, so the two share an encoding history. Whether codec, resolution or bitrate alone separate controls is tested on a prespecified subset, not assumed. False-positive rates are estimated on independent real scenes; correlated windows of one recording are not independent scenes. For example, zero false positives in 30 independent negative scenes still leaves a one-sided 95% upper bound near 9.5%. Measures that fail remain diagnostics.

B. Human Alignment

Independent source-role annotations and blinded human judgments are required. Following common practice for generated-video benchmarks [3], [4], model-level agreement is reported as the Spearman correlation between metric-derived and human-derived win ratios per dimension. Following quality-assessment practice [25], [26], video-level agreement is reported as SRCC, PLCC and KRCC against mean opinion scores. Inter-rater reliability is reported as Krippendorff's α [31]. Raters answer separate questions on support stability and integrity, material presence and location, amount of activity, motion plausibility, direction, and replay versus legitimate periodicity, each with *cannot tell* and *not applicable* options. Model-level win ratios, video-level opinion scores and paired preferences are analysed separately. A source-conditioned real-versus-generated choice is reported as a further discrimination test; because sharpness and codec cues can drive it, it does not establish physical validity. [Pending: human-study results, annotator counts and protocol]

C. Incremental Validity and Robustness

Each admitted measure is compared with the closest established measure on the same videos and controls [3], [5], [7], [23], [24]: agreement is reported, and so is any case where the new measure adds nothing. Incumbents are held to the same admission rules as the new measures, and an incumbent that passes a control is reported as passing it. Rankings are recomputed under seed, frame-sampling, encoder and mask-erosion perturbations, and their stability is reported as rank correlation with intervals. [Pending: robustness results]

VI. Benchmark Results

A. Existing Evidence

The existing source-fixed port was checked on two nine-scene, 60-second subsets against stored scores. These checks establish numerical reproduction on those inputs, not construct validity. Existing 60-second image-conditioned scores permit matched comparison of source-fixed SNF, VBench-I2V and VBench-Long on shared source scenes. The 120- and 240-second method records score 60-second endpoint windows of separately requested generations; these should not be interpreted as trajectories of a single rollout. Historical text-to-video records use different mask and measurement identities and are reported separately.

Figure 4 shows eight historical text-to-video checkpoints at the longest existing horizon; Fig. 5 shows all four prompts with outputs from all eight, with none excluded. Seven are historical outputs from the recorded common inference configuration and Steady-Forcing [29] is its released checkpoint under its native configuration, all on the earlier prompt set rather than the expanded suite. A source-matched image-to-video panel awaits native renders of the published entrants.

B. Leaderboard

Results are reported per track and horizon as per-axis profiles with 95% source-clustered bootstrap intervals [35],

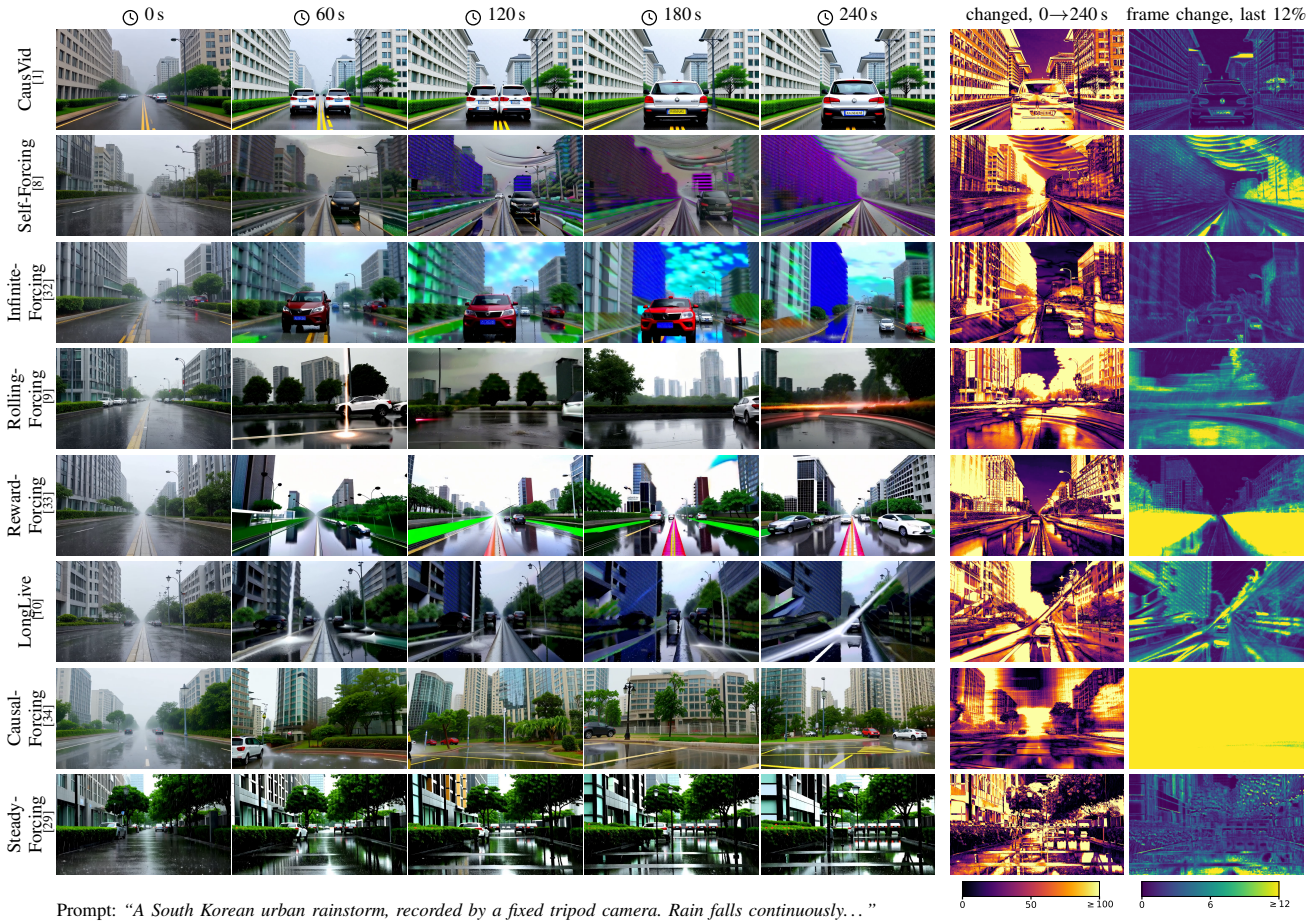

Fig. 4. **Eight released checkpoints over a nominal 240 s fixed-camera rollout.** One historical output per checkpoint at 832×480 (lossless extracted frames): the first seven under the common inference configuration, Steady-Forcing under its native configuration. The main prompt is the first by prompt ID of the four 240 s prompts for which all eight T2V checkpoints have an output (all four: Fig. 5). Right: grey-level change between the mean of the first and the last second, and mean successive-frame change in the last 12% at 8 fps, on fixed colour scales; these are visualizations without region masks, not metrics. Changes include both intended motion and scene changes; they do not by themselves distinguish camera motion, flicker or redrawing. The 240 s label denotes the final available frame, slightly before 240 s for these files. One output per checkpoint on one prompt cannot rank systems. Interactive time-synchronized viewer: `interactive/qualitative_viewer.html`.

macro-averaged over regimes, with eligibility counts beside every value. No single summed score is reported. [Pending: leaderboard tables per track and horizon]

C. Per-Regime Analysis and Comparison with Established Metrics

[Pending: per-regime breakdown; agreement with VBench and WorldScore dimensions]

[Pending: rights-cleared independent confirmation set, independently reviewed role maps, native long-video baselines, sufficient multi-seed coverage, real-periodicity controls and human construct validation. Previously scored videos and metrics are reused rather than regenerated or rescored where their input and protocol identity is verified.]

VII. Findings and Discussion

[Pending: findings supported by the confirmation results, each with its interval and eligible population. Candidate questions: does motion persist or decay over the horizon; does support fidelity trade against material activity; do regimes differ in difficulty; where do established metrics disagree with the measures here.]

VIII. Limitations, Ethics and Reproducibility

Existing result populations have development exposure, and passing numerical parity does not establish perceptual validity. Source-relative geometry, activity and recurrence can each fail under plausible counterexamples. Real-reference bands depend on the real-footage set: they are reported with that set's size and medium coverage and are recomputed, not edited, when it grows.

Any final release must include exact input and model provenance, metric versions, failure and abstention counts, annotation uncertainty and rights-cleared material. Every source, real clip and generated video is released only under a recorded licence; an asset without one is withheld. The release follows a datasheet [36] (Appendix B) and versions both the inputs and the measurement identities, so later results are never pooled with earlier ones. Human-study procedures, consent and compensation are reported with the study. [Pending: maintenance plan and hosting]

Publication overlap with the original SNF-Bench [13], SceneTether [30] and Steady-Forcing [29] work must be disclosed. Metric and threshold selection do not use the standing of either author-affiliated method, and results are reported with and without those methods.

IX. Conclusion

The working protocol treats long-horizon fixed-camera video evaluation as a multi-axis measurement problem in which each statistic is read two-sided against real footage and admitted only after adversarial and human validation. Evidence for reliable naturalness assessment awaits the independent experiments and human study specified above.

Appendix A
Relation to the SNF-Bench Preprint

The SNF-Bench preprint [13] introduced region-separated drift and flow measures on an earlier prompt set. This manuscript differs in its rebuilt and contamination-audited suites, source-derived role maps, two-sided realism measures, adversarial admission protocol, human alignment and entrant panel. No earlier result is pooled with the present measurements. [Pending: itemized new-content statement with proportions, as journal policy requires]

Appendix B
Datasheet Summary

[Pending: motivation, composition, collection process, preprocessing, uses, distribution and maintenance, following [36]]

Appendix C
Qualitative Outputs at 240 Seconds

Figure 5 extends Fig. 4 to every qualifying prompt.

Appendix D
Known-Answer Development Results

Figure 6 reports how two development candidates respond to constructed controls with known answers. These are development data from one model family and are not admission evidence.

Appendix E
Measurement Definitions

[Pending: exact definition, eligibility rule, abstention rule and identity string of every admitted measure]

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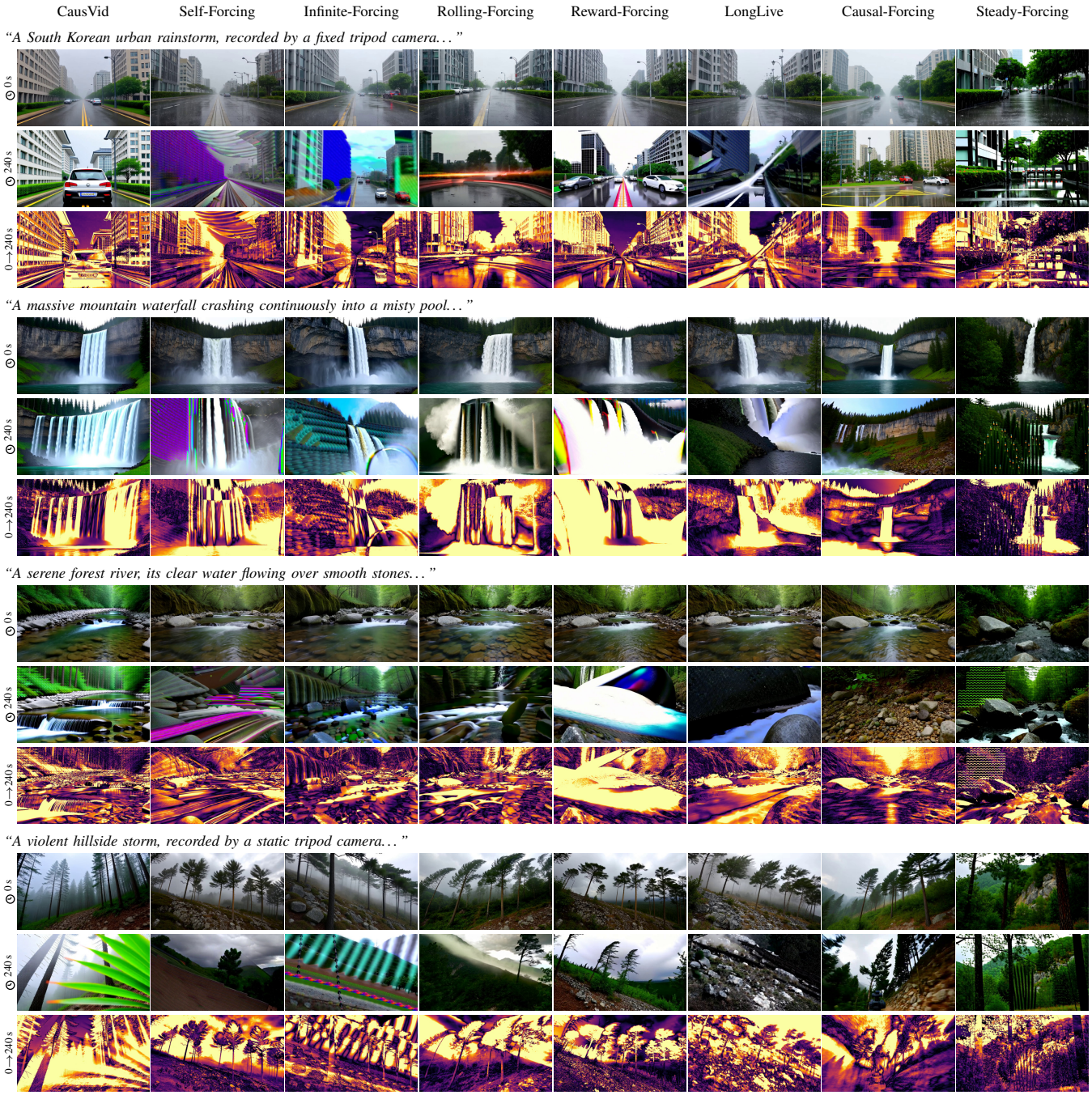

Fig. 5. **All four nominal 240 s prompts with outputs from the eight T2V checkpoints** (the first seven under the common inference configuration, Steady-Forcing under its native one). First and last frame and the 0→240 s change map (colour scale as in Fig. 4) of one output per system and prompt; no output was excluded. Columns are systems in registry order, not ranks. Late-motion maps and frames every 10 s are in `interactive/qualitative_viewer.html`.

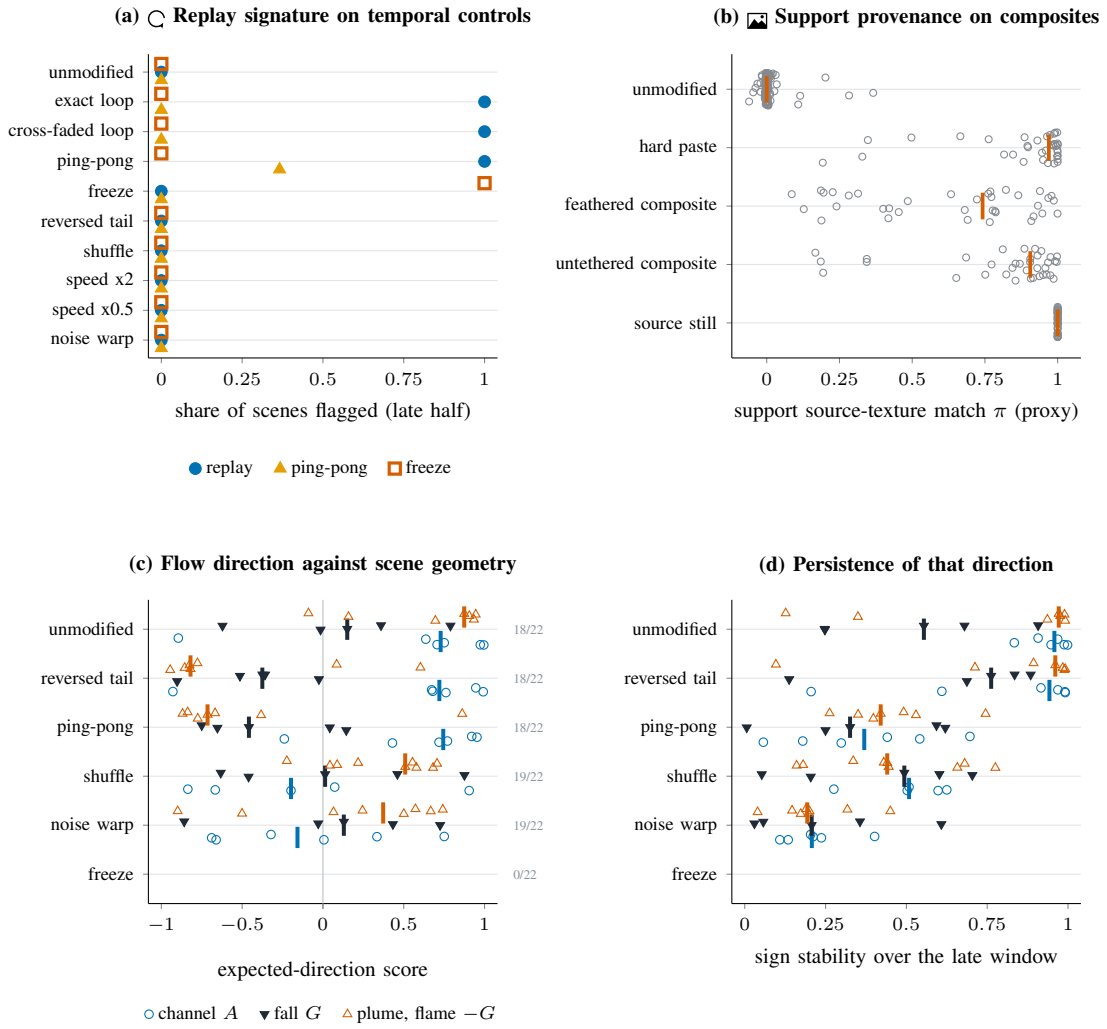

Fig. 6. **Known-answer responses of development candidates** (one model family, one seed, constructed controls; no real footage, no human labels; separability, not validity). (a) `cand-replay-signature-0`, the repetition diagnostic of the authors' method project [30] run on SNF role maps, on 41 scenes per control: unmodified outputs and time-reversed, shuffled, re-timed and noise-warped controls are never flagged as replay; loops and ping-pong are; freezes are flagged as frozen. (b) Source-texture match of the support (normalized correlation of high-pass residuals with the source, relative to a codec still) for unmodified outputs (three cells, 108 videos) and three composite constructions (36 each); bars: medians. Composites separate from unmodified outputs (AUROC 0.996–1.000), but where the line falls depends on the model family, so no threshold is set before the decoder-range ceiling is measured. (c) Direction referenced to the scene, not the caption, on the 22 routed scenes (6 channels, 5 falls, 11 plumes and flames) at the frozen-spec rate: axial alignment A of the material flow with the bank tangent for channels (sign-free, $+1$ along, -1 across), gravity sign G for falls and $-G$ for plumes and flames ($+1$ = the physically expected way). Right: scenes scored/attempted; the rest abstain because the material moves no more than its support. Reversal flips G but leaves A unchanged, as designed; every freeze abstains. Falls are the weak route at this rate (median G 0.15, 3/5 downward); sampling every frame raises it to 0.37 (4/5), consistent with fast water outrunning the flow estimator at 8 fps. Bars: medians per route. (d) Sign stability of that direction over the late window: near 1 for channels, plumes and flames on unmodified and reversed tails (falls 0.55 and 0.76), and 0.2–0.5 under ping-pong, shuffle and noise warp, which a time-averaged direction alone cannot separate.